

A5. Kompyuterning arifmetik va mantiqiy asoslari

Maqsad:

Talabalarga kompyuterlarning arifmetik va mantiqiy ishlash tamoyillarini tushuntirish, mantiqiy amallar va arifmetik-mantiqiy qurilma (ALU) faoliyati haqida amaliy tushuncha berish.

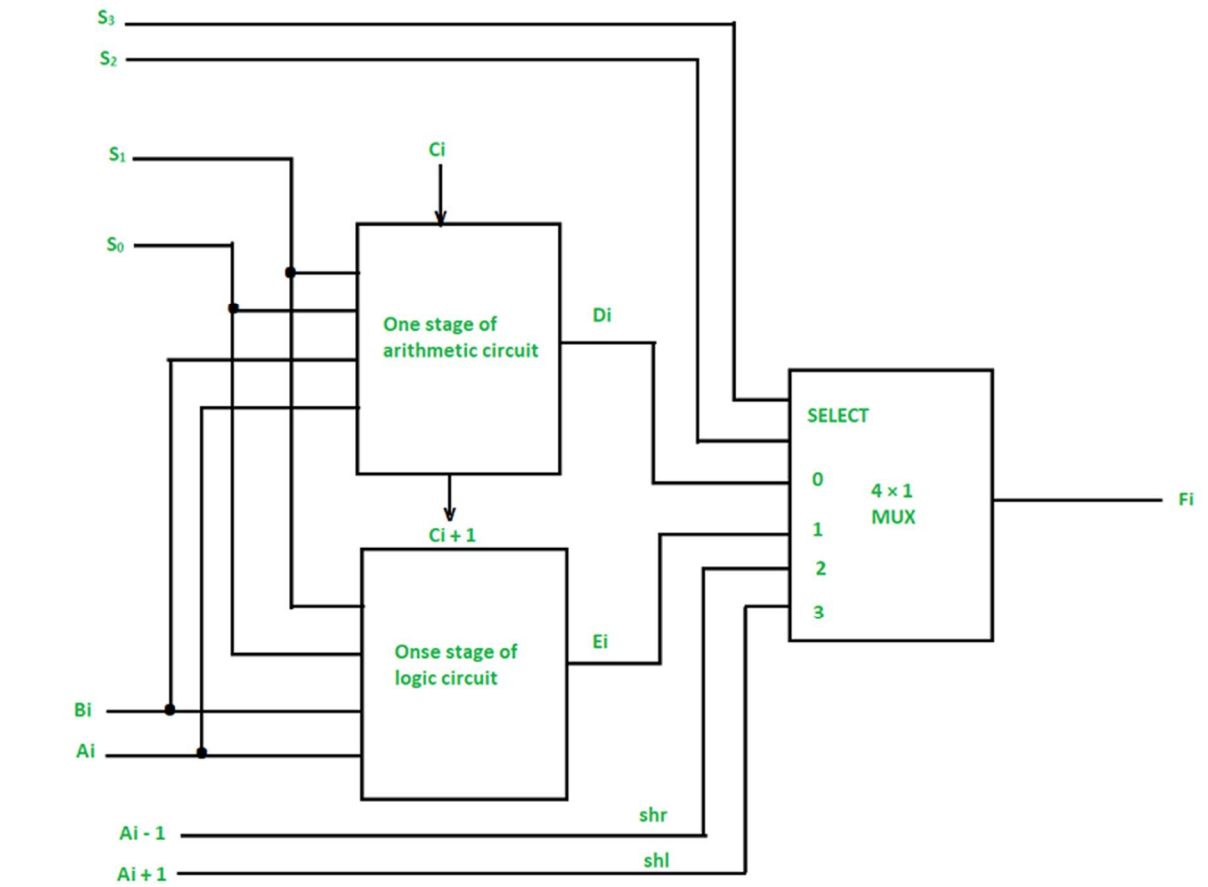
Vazifalar:

1. Arifmetik va mantiqiy amallar tushunchasini o'rganish
2. Mantiqiy elementlar (AND, OR, NOT va boshqalar) bilan tanishish
3. Arifmetik-mantiqiy qurilma (ALU) vazifalarini tushunish
4. Mantiqiy amallar asosida misollar yechish

Nazariy qism

1. Kompyuterning arifmetik va mantiqiy asoslari tushunchasi

Kompyuter ma'lumotlarni qayta ishlash jarayonida **arifmetik** (qo'shish, ayirish, ko'paytirish, bo'lish) va **mantiqiy** (taqqoslash, tekshirish, shartli tanlash) amallardan foydalanadi. Ushbu amallar asosan **ikkilik sanoq tizimi** asosida bajariladi.



		y	
	$\wedge$	0	1
x	0	0	0
	1	0	1

		y	
	$\vee$	0	1
x	0	0	1
	1	1	1

		y	
	$\rightarrow$	0	1
x	0	1	1
	1	0	1

		y	
	$\oplus$	0	1
x	0	0	1
	1	1	0

Figure 1. Truth tables

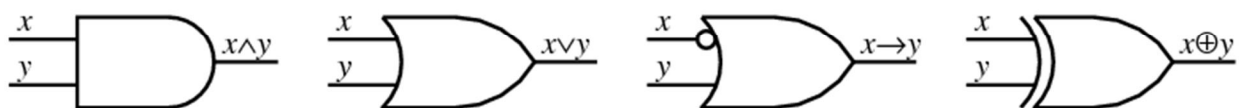


Figure 2. Logic gates

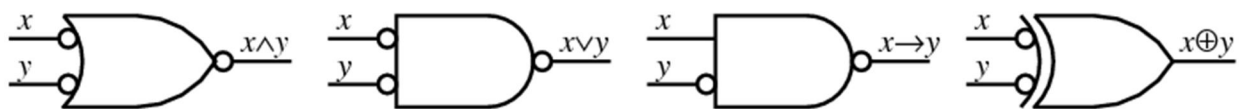


Figure 3. De Morgan equivalents



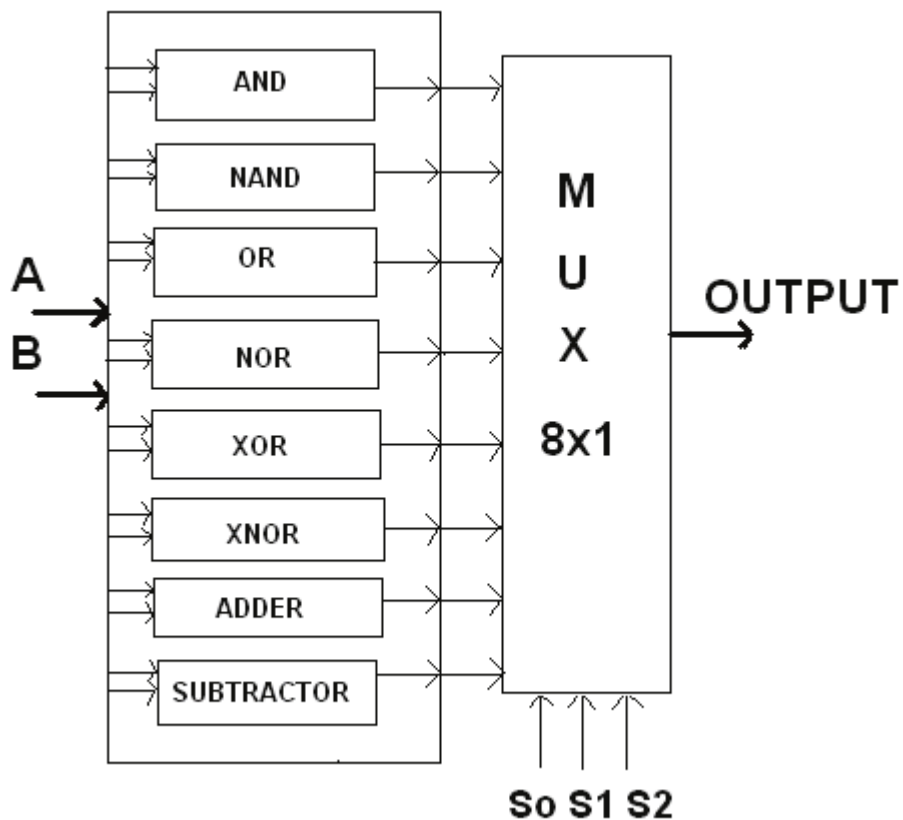
Figure 4. Venn diagrams

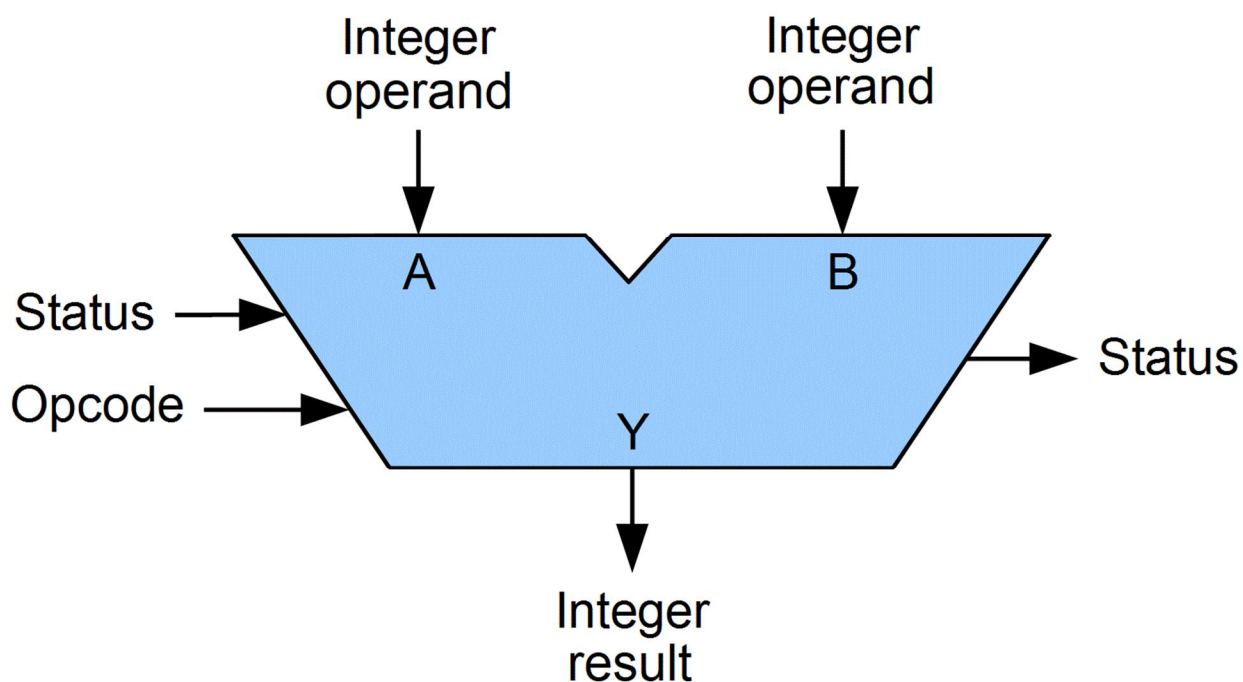
2. Arifmetik-mantiqiy qurilma (ALU)

ALU — bu markaziy protsessorning asosiy qismi bo'lib, barcha arifmetik va mantiqiy hisob-kitoblarni bajaradi.

ALU bajaradigan amallar:

- Qo'shish (+)
- Ayirish (−)
- Ko'paytirish (×)
- Bo'lish (÷)
- Mantiqiy AND, OR, NOT, XOR
- Taqqoslash (>, <, =)





3. Mantiqiy amallar va ularning turlari

a) AND (VA) amali

Natija 1 bo'lad, agar **ikkala kirish ham 1 bo'lsa**.

A B A AND B

0 0 0

0 1 0

1 0 0

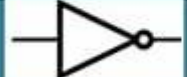
1 1 1

Input AND Gate

TRUTH TABLE

INPUTS		OUTPUT
X	Y	Z
0	0	0
0	1	0
1	0	0
1	1	1

$Z = XY$

		AND
		OR
		NOT
		NAND
		NOR
		XOR
		XNOR

b) OR (YOKI) amali

Natija 1 bo'ladi, agar **kamida bittasi 1 bo'lsa**.

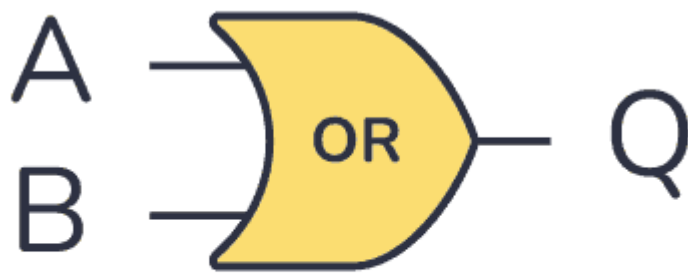
A B A OR B

0 0 0

0 1 1

1 0 1

1 1 1



A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

c) NOT (EMAS) amali

Kirish qiymatini **teskari** qiladi.

A NOT A

0 1

1 0



A	Q
0	1
1	0

d) XOR (Istisno YOKI)

Natija 1 bo'ladi, agar **kirishlar turli bo'lsa**.

A B A XOR B

0 0 0

0 1 1

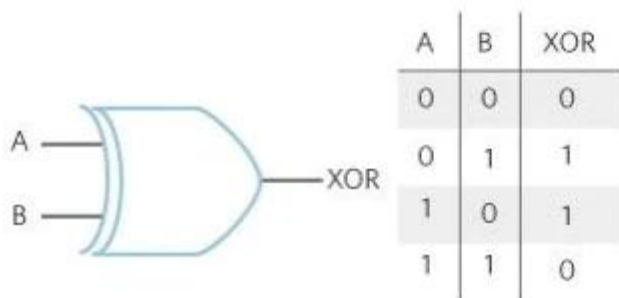
1 0 1

1 1 0



A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

$$X = A \oplus B$$



4. Mantiqiy amallarning kompyuterdagi qo'llanilishi

- Shartli operatorlar (if, else)
- Taqqoslash amallari
- Bitli amallar (bitwise operations)
- Xavfsizlik va shifrlash tizimlari
- Protessor boshqaruv signallari

Operator	Description
&	bitwise AND
	bitwise OR
^	bitwise exclusive OR
<<	shift left
>>	shift right
~	one's complement

C_1	C_0	Arithmetic $C_2 = 0$	Logical $C_2 = 1$
0	0	Addition	OR
0	1	Subtraction	AND
1	0	Multiplication	NOT
1	1	Division	EX-OR

Amaliy topshiriqlar

- Quyidagi mantiqiy amallarni bajaring:
 - 1 AND 0
 - 1 OR 0
 - NOT 1
 - 1 XOR 1
- Ikkilik ko'rinishda hisoblang:
 - 1011_2 AND 1101_2
 - 1010_2 OR 0110_2
- ALU ning kompyuterdagi rolini izohlang.
- Quyidagi ifodaning natijasini toping:
(1 AND 0) OR (NOT 0)

Xulosa

Mazkur amaliy ish orqali talabalar kompyuterning arifmetik va mantiqiy asoslarini, ALU ning ishlash tamoyillarini hamda mantiqiy amallarning amaliy ahamiyatini chuqur o'rganadilar. Bu bilimlar dasturlash, raqamli elektronika va kompyuter arxitekturasi fanlari uchun asos bo'lib xizmat qiladi.

Nazorat savollari

1. Kompyuterning arifmetik asoslari deganda nima tushuniladi va u qanday amallarni o'z ichiga oladi?
2. Mantiqiy amallar nima va ular kompyuter tizimlarida qaysi maqsadlarda qo'llaniladi?
3. Arifmetik-mantiqiy qurilma (ALU) nima? ALU ning asosiy vazifalarini tushuntirib bering.
4. AND, OR, NOT va XOR mantiqiy amallarining ishlash prinsipini izohlang.
5. Mantiqiy amallar dasturlash va protsessor ishlashida qanday ahamiyatga ega? Misol bilan tushuntiring.